

Roll No.

TMA-201

B. TECH. (SECOND SEMESTER)
MID SEMESTER EXAMINATION, March, 2024
ENGINEERING MATHEMATICS—II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Solve the differential equation $x^2 y dx - (x^3 + y^3) dy = 0$. (CO1)

OR

- (b) Define linear differential equation with constant coefficient. Solve the following differential equation : $D^2 y + Dy - 2y = x + \sin x$. (CO1)

2. (a) Solve by Variation of parameter : $\frac{d^2 y}{dx^2} + a^2 y = \sec ax$. (CO1)

OR

- (b) Solve the equation : $\frac{d^4 y}{dx^4} - y = \cos x$. (CO1)

3. (a) Find Laplace transform of $\frac{(e^{at} - \cos bt)}{t}$. (CO2)

P. T. O.

(2)

OR

(b) Define Unit Step Function. Find the Laplace of the given function :

$$(t - 3)^2 u(t - 3).$$

(CO2)

4. (a) State Convolution theorem. Use Convolution theorem to find : (CO2)

$$L^{-1} \left(\frac{p^2}{(p^2 + a^2)(p^2 + b^2)} \right).$$

OR

(b) Solve by using Laplace Transform :

$$\frac{d^2 y}{dt^2} - 3 \frac{dy}{dt} + 2y = 4t + e^{3t}$$

$$\text{where } y(0) = 1, y'(0) = -1.$$

(CO2)

5. (a) Solve the equation :

(CO1)

$$x^2 \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} + 5y = x^2 \sin(\log x).$$

OR

(b) Find the value of $\int_0^\infty e^{-3t} t \sin t \, dt$.

(CO2)

Roll No.

TPH-201

B. TECH. (SECOND SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

ENGINEERING PHYSICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) In Fresnel biprism experiment deduce, the expression to determine the distance between two virtual sources, using deviation method and displacement method. (CO1)

OR

- (b) In Newton's ring experiment, derive the expression for diameter of bright rings when a liquid of refractive index μ is placed between the lens and the plate. Explain why broad source of light is used in interference in thin films. (CO1)
2. (a) Derive the expression for path difference in a wedge shape film in reflected light. Is the point of contact between lens and plate bright or dark in Newton's ring experiment ? Give reason. (CO1)

P. T. O.

OR

- (b) Light source emitting wavelength $\lambda_1 = 6000 \text{ \AA}$ and $\lambda_2 = 4800 \text{ \AA}$ is used to obtain Newton's rings in reflected light. It is found that n th dark ring due to λ_1 coincides with $(n + 1)$ th dark ring due to λ_2 . If the radius of curvature of the curved surface of the lens is 0.96 m, calculate the diameter of $(n + 1)$ th dark ring of λ_2 . (CO1)

3. (a) Derive the expression for intensity distribution due to Fraunhofer diffraction at single slit. Find the ratio between the intensities of successive maxima's. (CO1)

OR

- (b) A plane transmission grating having 5000 lines/cm is used to obtain a spectrum of light from sodium lamp in third order. Calculate the angular separation between the two sodium lines whose wavelength are 5890 Å and 5896 Å. (CO1)

4. (a) Explain the term modes of propagation in an optical fiber. Give differences between single mode step index fiber and multimode graded index fiber. (CO2)

OR

- (b) In an optical fiber, the refractive index of core is 1.5 and cladding is 1.48. Find the critical angle, fractional change in refractive index, numerical aperture and acceptance angle. (CO2)

(3)

5. (a) Write the working principle of an optical fiber. Also derive the expression for acceptance angle and numerical aperture. (CO2)

OR

- (b) Calculate the refractive index of the core and cladding material of a fiber having Numerical Aperture = 0.22 and fractional refractive index change $(\Delta\mu_r) = 0.012$. (CO2)

(b) Write the winning principle of an ethical idea. Also define the expectation for acceptance and ethical actions.

(c) Calculate the relative index of the two and finding pattern of the first having numerical amount 0.2 and fractional relationship.

Roll No.

TCH-201

B. TECH. (SECOND SEMESTER)
MID SEMESTER EXAMINATION, March, 2024
ENGINEERING CHEMISTRY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Using molecular orbital theory, compare the bond energy and magnetic character of O_2 , O_2^+ and O_2^- species. Calculate the bond order of each.

(CO1)

OR

- (b) Discuss 5 postulates of Molecular Orbital Theory. With respect to Band Theory, explain why sodium metal is a good conductor of electricity while wood is an insulator.

(CO1)

2. (a) Differentiate between intermolecular and intramolecular hydrogen bonding. State 5 unique properties of water.

(CO1)

OR

- (b) Explain different types of transitions involved in UV Spectroscopy. Write applications of UV Spectroscopy.

(CO1)

P. T. O.

(2)

3. (a) What is zeolite. Explain zeolite process for water softening with its benefits, and limitations. (CO2)

OR

- (b) Calculate the amount of lime (85% pure) and soda (90% pure) required for treatment of 20,000 litres of water containing : $\text{Mg}(\text{HCO}_3)_2 = 18.2$ ppm, $\text{CaCl}_2 = 13.8$ ppm, $\text{Ca}(\text{HCO}_3)_2 = 20.3$ ppm, $\text{MgCl}_2 = 17.0$ ppm, $\text{MgSO}_4 = 15.0$ ppm, $\text{CaSO}_4 = 25.0$ ppm and $\text{NaCl} = 25$ ppm. (CO2)
4. (a) Write note on the following : (CO2)
- (i) Reverse osmosis
 - (ii) Disadvantages of use of hardwater

OR

- (b) A completely exhausted zeolite softener requires 80 liters of NaCl solution having 100 g/l of NaCl . Calculate the hardness of water if 1500 liters of water was softened by the zeolite softener. (CO2)
5. (a) Explain lime soda process used for softening of hard water by giving suitable reactions. What is the use of coagulants in lime soda process ? (CO2)

OR

- (b) What are the advantages and disadvantages of use of hard water ? Two water samples contain 408 mg of CaSO_4 per liter (solution-1) and 408 mg of CaCl_2 per liter (solution-2). Express their hardness in CaCO_3 equivalents and state which solution has a higher hardness. (CO2)

Roll No.

TCS-201

B. TECH. (SECOND SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

PROGRAMMING FOR PROBLEM SOLVING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) All parts of Q. No. 2 are compulsory.

(iii) Each sub-question carries 10 marks.

1. (a) How 2D array is different and similar from/to 1D array ? What do you mean by row major and column major ? Consider a float type 2D array Arr[80][100]. The base address of array is 2000. Calculate the address of the element Arr[45][55] using both row major and column major.

(CO1)

OR

- (b) Write a C code to input elements in a matrix of size $r \times c$. Find minimum element of each column.

(CO1)

P. T. O.

2. Find the output with explanation :

(CO1, CO3)

(1) #include<stdio.h>

```
void main()
```

```
{
```

```
int a[5]={1,2,3,4,5},i;
```

```
int *p=a+3;
```

```
printf("%d %d\n", *p, p-a);
```

```
}
```

(2) #include<stdio.h>

```
void main()
```

```
{
```

```
char str[] = {'a', 'b', '\0', 'c', 'd', '\0'};
```

```
printf("%d %d", sizeof(str);
```

```
strlen(str));
```

```
}
```

(3) #include<stdio.h>

```
void main()
```

```
{
```

```
int A[10]={1,2,3,4};
```

```
static int a, b, i;
```

```
for(i=0;i<10;++i)
```

```
{
```

```
if(A[i]%2==0 && A[i]%5==0)
```

```
++a;
```

```
else
```

```
++b;
```

```
}
```

```
printf("a=%d,b=%d,a-b=%d\n",a,b,a-b);
```

```
}
```


(4) #include <stdio.h>

void main()

{

char s[20]="Graphic Era";

int l, i, j, t;

l=strlen(s);

for(i=0,j=l;i<j;i++,j--)

{

t=s[i];

s[i]=s[j];

s[j]=t;

}

printf("The string is: %s\n",s);

}

(5) #include <stdio.h>

void main()

{

int a[3] = {10, 20, 40};

int b[3];

b = a;

printf("%d", b[0]);

}

3. (a) Draw a flowchart to pass a 1D array of n different integer elements into a function and function should return the second smallest element of the array. (CO2)

OR

- (b) Draw a flowchart to pass a string into a function. Remove spaces from the string and print the final string. (CO2)
4. (a) Explain and differentiate with key features the different ways to read a string from keyboard. Write a C code to input a string, pass it to a function and check the string is palindrome or not, (using string library functions). (CO1)

OR

- (b) Write a C code to find number of times a substring appears in the string without using string library function : (CO1)

Sample Input	Sample Output
Main String : the cat is on the mat beside the hat Substring : the	Number of substrings appear : 3

5. (a) Define pointer and its type. What is the usage of pointer in C language ? (CO3)

OR

- (b) Write a C code to input n (where n is even) numbers in an array than by passing this array to a function that interchange element at odd index with elements at even index (access array in the function using pointer) : (CO3)

Sample Input : 2 4 6 7 1 3

Sample Output : 4 2 7 6 3 1

Roll No.

TEE-201

B. TECH. (SECOND SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

BASIC ELECTRICAL ENGINEERING

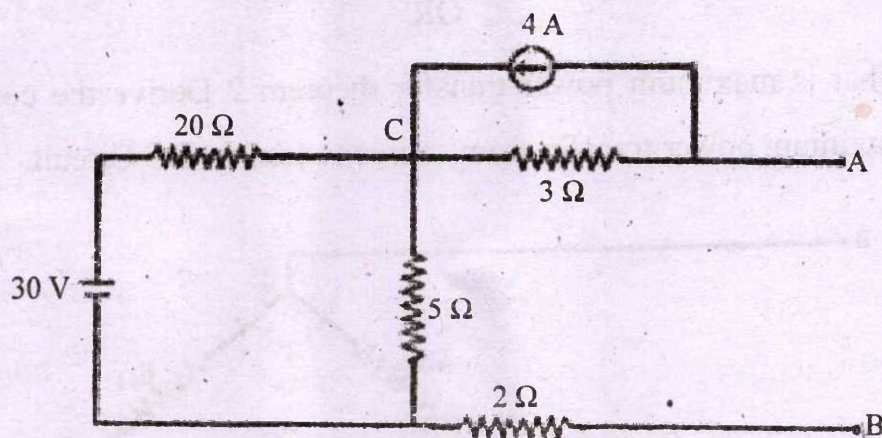
Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a)



State Thevenin theorem and find the Thevenin equivalent circuit for the above circuit. (CO1)

OR

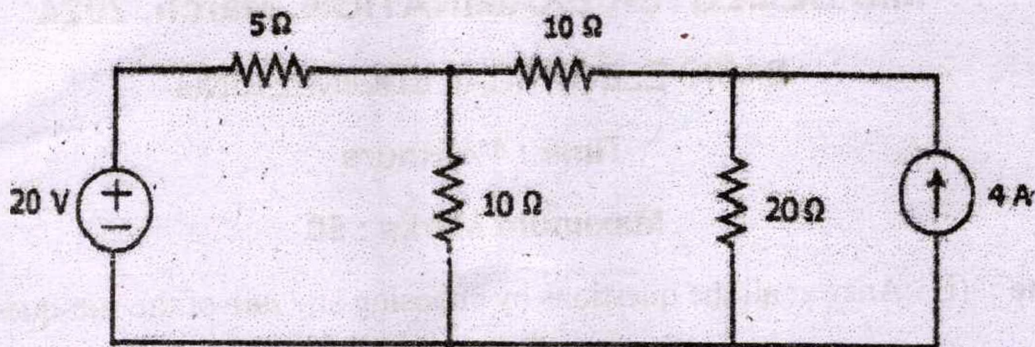
(b) Define the following terms along with suitable examples : (CO1)

(i) Active and Passive Elements

P. T. O.

- (ii) Unilateral and Bilateral Elements
- (iii) Supernode and Supermesh condition
- (iv) Open circuit and Short circuit
- (v) Ideal and practical sources

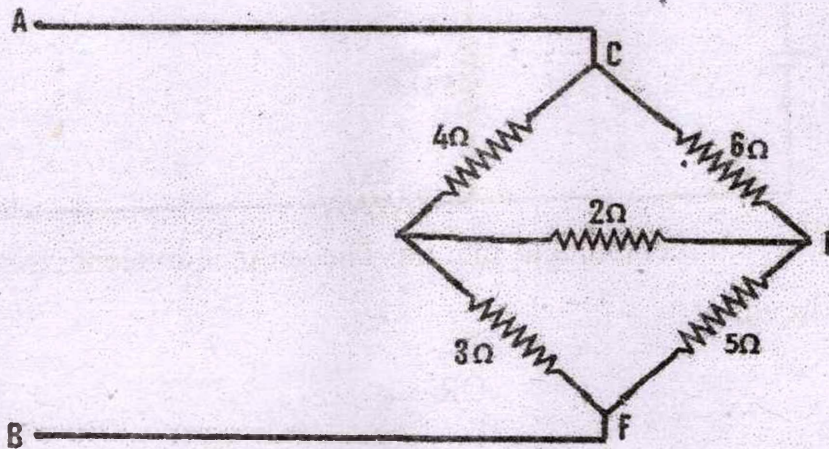
2. (a) Calculate current through 20Ω resistor using Superposition theorem in the circuit shown in the following figure. (CO3)



OR

- (b) What is maximum power transfer theorem ? Derive the condition for maximum power transfer from source to load in DC Circuit. (CO3)

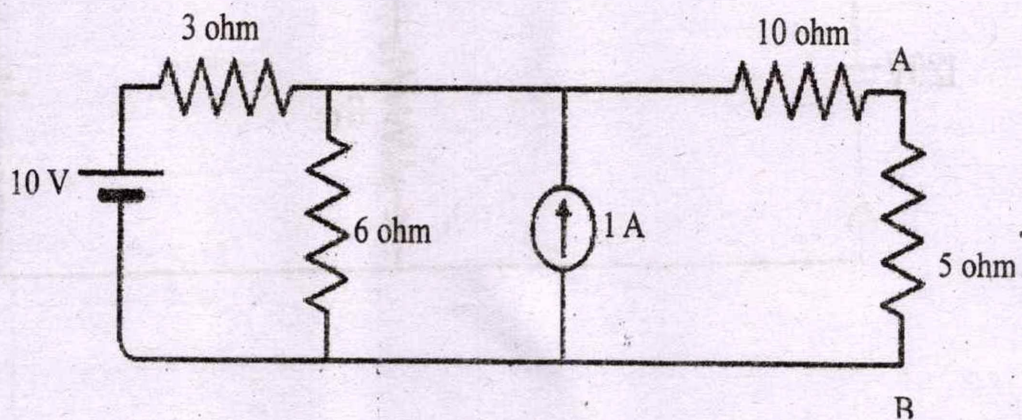
3. (a)



Find the resistance between terminals AB.

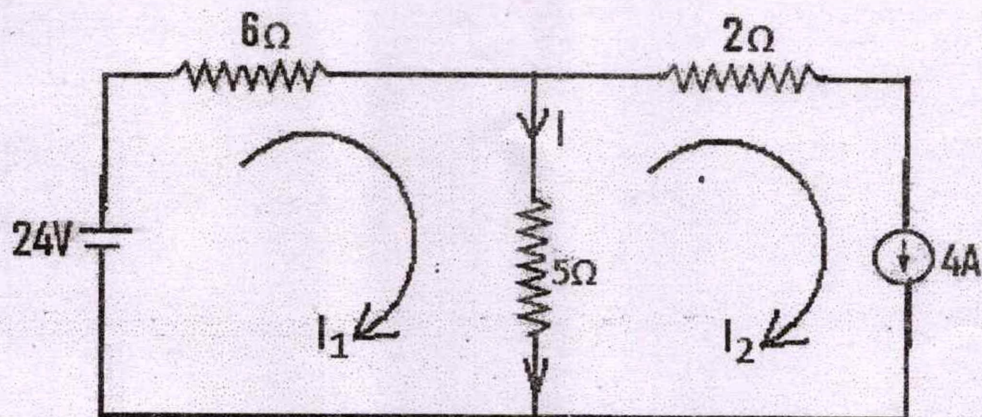
OR

- (b) What are Star and Delta configurations ? Derive expression for Delta to Star transformation.
4. (a) State the Norton's theorem. Find out current in 5Ω resistor with the help of Norton's Theorem in the circuit shown in the figure below.



OR

- (b) Find the current I in the circuit of figure using Mesh Analysis. (CO2)



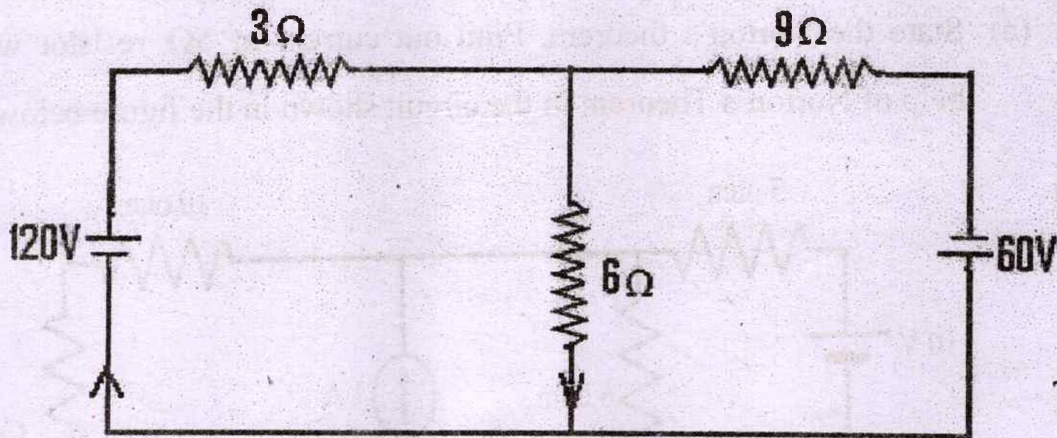
5. (a) Derive the formulae for current and voltage division rule. (CO2)

(4)

TEE-201

OR

- (b) In the circuit shown in figure find the current through 6 ohm resistor by using Nodal Analysis.



TEE-201

600

Roll No.

TEC-201

B. TECH. (SECOND SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

BASIC ELECTRONICS ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) (i) $(A9BC2.0C)_{16} = (?)_{10} = (?)_6$. (CO1)

(ii) $(432)_x = (225)_7$, find the base 'x'.

OR

(b) Simplify the following expressions with the help of Boolean algebra :

(CO1)

(i) $F = (AB)' + A(B'C') + B(B + C)'$

(ii) $F = [(AB)'(C + BD) + A'B']C$

(iii) $A'BC + AB'C' + A'B'C' + AB'C + ABC$

(iv) $F = A'BC + AB'C' + A'B'C' + AB'C + ABC$

(v) $F = AB + (AC)' + AB'C(AB + C)$

P. T. O.

2. (a) Simplify the following function with the help of K-map : (CO1)

$$F(A, B, C, D) = A + B'C' + A'BD' + AB'C'D$$

OR

- (b) Minimize the following function with the help of K-map and realize the minimized function by NAND gate only as well as NOR gate only :

(CO1)

$$F(A, B, C, D) = \sum m(1, 3, 4, 6, 8, 9, 11, 13, 15) + \sum d(0, 2, 14)$$

3. (a) If N_D and N_A are donor and acceptor impurities and n_i is the intrinsic concentration, establish the relation for minority and majority charge densities in extrinsic semiconductors. (CO2)

OR

- (b) Derive an expression for drift and diffusion current. (CO2)

4. (a) A sample of Ge is made p -type by adding acceptor atoms at a rate of one atom per $4.5 \times 10^8/\text{m}^3$. (CO2)

If density of Ge atoms is $3.4 \times 10^{28}/\text{m}^3$, determine electron and hole concentration in the doped semiconductor, if $n_i = 2.5 \times 10^{19}/\text{m}^3$.

OR

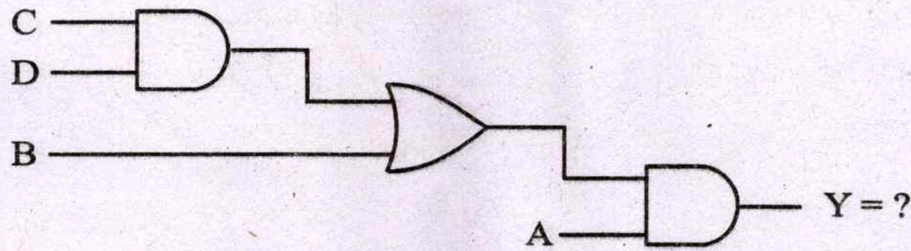
- (b) Calculate the current produced in a Ge plate of area 1.5 cm^2 and thickness 0.4 mm when a potential difference of 2.5 V is applied across the faces. Given that the concentration of free electrons is $2 \times 10^{19}/\text{m}^3$ and mobility of electrons and holes are $0.25 \text{ m}^2/\text{V-sec}$ and $0.07 \text{ m}^2/\text{V-sec}$ respectively. (CO2)

(3)

5. (a) A bar of pure silicon has cross-sectional area of 1 mm^2 and intrinsic concentration of silicon is $2.5 \times 10^{16} \text{ m}^{-3}$. The free electron and hole mobilities are $0.23 \text{ m}^2/\text{V-sec}$ and $0.15 \text{ m}^2/\text{V-sec}$ respectively. Find the conductivity and length of bar whose resistance is $100 \text{ k}\Omega$. (CO1/CO2)

OR

- (b) Define the output 'Y' of the logic circuit given below and write the truth table for the same. (CO1/CO2)



(a) A bar of pure silicon has cross-sectional area of 1 mm^2 and length 10 cm . The electron concentration is $1.5 \times 10^{16} \text{ m}^{-3}$. The hole concentration and the resistivity are 0.15 m^{-3} and 2.1 m^{-3} respectively. Find the conductivity and length of bar whose resistance is $100 \text{ k}\Omega$. (601100)

(b) Label the output Y of the logic circuit given below and write the truth table for the same. (601100)



Roll No.

TCE-201

**B. TECH. (SECOND SEMESTER)
MID SEMESTER EXAMINATION, March, 2024**

BASIC CIVIL ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Write a short note on the history of Civil Engineering. (CO1)

OR

- (b) Explain the role of civil engineers in infrastructural development of nation. (CO1)

2. (a) Describe the following structures : (CO1)

(i) Tunnels

(ii) Retaining walls

(iii) Dams

(iv) Road

OR

- (b) Enlist the various branches of Civil Engineering and briefly explain any *two*. (CO1)

P. T. O.

(2)

3. (a) Distinguish between plans and maps. Define representative fraction.

(CO2)

OR

- (b) Discuss the main principles of surveying. Also explain them with the help of neat sketch.

(CO2)

4. (a) Convert WCB into QCB for the following :

(CO2)

(i) 193°

(ii) 239°

OR

- (b) Write short notes on 'Geographical Information System' and 'Global Positioning System'.

(CO2)

5. (a) Convert QCB into WCB for the following :

(CO2)

(i) $S 22^\circ W$

(ii) $N 61^\circ W$

OR

- (b) Write a short note on "Various Area of Application of Remote Sensing" and how it is integrated with GIS.

(CO2)